IMAGING COURSE · LESSON 13

Lymph nodes

Every head and neck scan is also a scan of the lymph nodes, and the nodes often matter more than the primary: nodal metastasis is the single most important prognostic factor in head and neck squamous carcinoma, and a node is how many cancers first present. This lesson teaches the imaging levels from their landmarks, the features that make a node suspicious, the patterns that point to the primary, and the appearances of the non-cancerous causes you will meet in clinic.

AT A GLANCE

- Levels from landmarks: **hyoid** (II/III), **cricoid** (III/IV), **back of sternocleidomastoid** (V), **jugular vein** (IIA/IIB), **manubrium** (VI/VII).
- Malignant features together: **necrosis**, **round shape** and lost hilum, size, clustering, **extranodal extension**.
- Do not forget the **retropharyngeal** nodes: first station for the nasopharynx.
- A **cystic node** in an adult is metastasis until proven otherwise: HPV oropharynx or papillary thyroid.
- Carotid contact **≥270°** suggests encasement and unresectability.

1 CHAPTER ONE Levels from landmarks

Translating the neck dissection to the scan.

1.1 The landmarks

The surgical levels are drawn on imaging from fixed landmarks that can be found on any axial slice: the **mylohyoid** and the **posterior edge of the submandibular gland**, the **hyoid bone**, the lower edge of the **cricoid cartilage**, the **posterior edge of the sternocleidomastoid**, the **internal jugular vein**, the **carotid arteries** and the top of the **manubrium**.

IMAGING LEVELS

LEVEL	BOUNDARIES ON AXIAL IMAGING
IA	Submental: between the anterior bellies of the digastrics, below the mylohyoid

LEVEL	BOUNDARIES ON AXIAL IMAGING
IB	Submandibular: lateral to the anterior digastric, in front of the posterior edge of the submandibular gland
IIA	Skull base to the lower body of the hyoid, behind the submandibular gland, in front of the posterior edge of the sternocleidomastoid; in front of, beside or inseparable from the internal jugular vein
IIB	As IIA, but behind the internal jugular vein with a fat plane between
III	Lower hyoid to the lower cricoid, in front of the posterior edge of the sternocleidomastoid
IV	Lower cricoid to the clavicle, in front of the posterior edge of the sternocleidomastoid
VA	Behind the posterior edge of the sternocleidomastoid, skull base to the lower cricoid
VB	Behind the sternocleidomastoid, lower cricoid to the clavicle
VI	Between the carotids, hyoid to the manubrium (prelaryngeal, pretracheal, paratracheal)
VII	Between the carotids, below the top of the manubrium, above the brachiocephalic vein

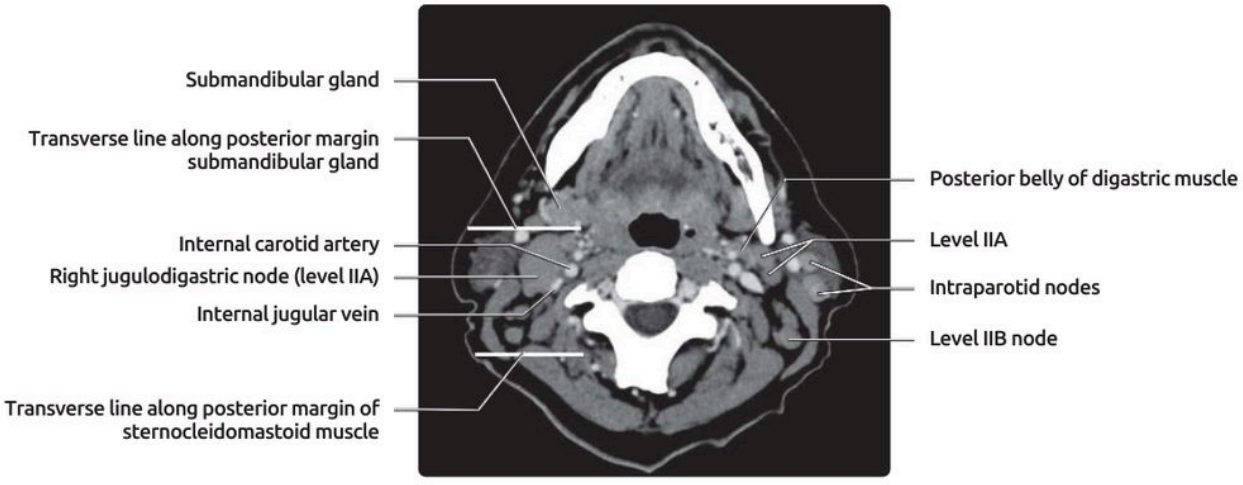


FIGURE 1 Axial contrast CT above the hyoid: lines along the posterior edges of the submandibular glands and sternocleidomastoids separate level I from level II, and level II from level V. The enlarged right level II node is the jugulodigastric node.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 430

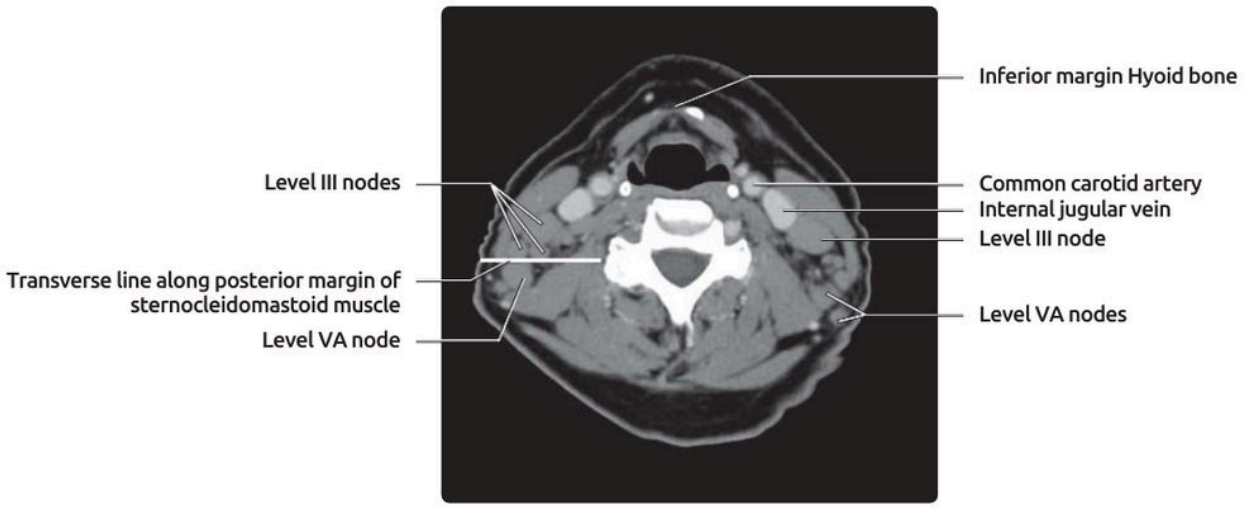


FIGURE 2 Just below the hyoid: level III nodes along the internal jugular vein in front of the line along the back of the sternocleidomastoid, level VA nodes behind it.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 431



FIGURE 3 At the top of the manubrium: the line between level VI above and level VII (the superior mediastinum) below.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 432

1.2 Nodes without a number

Some groups are named, not numbered, and are easily missed: the **retropharyngeal** nodes (lateral, or nodes of Rouvière, at the level of C1–C2 beside the longus capitis, and medial), the **parotid** and **facial** nodes, the **occipital** and **supraclavicular** nodes. Retropharyngeal nodes cannot be felt and are the first station for the nasopharynx and a common one for the oropharynx, hypopharynx and thyroid.

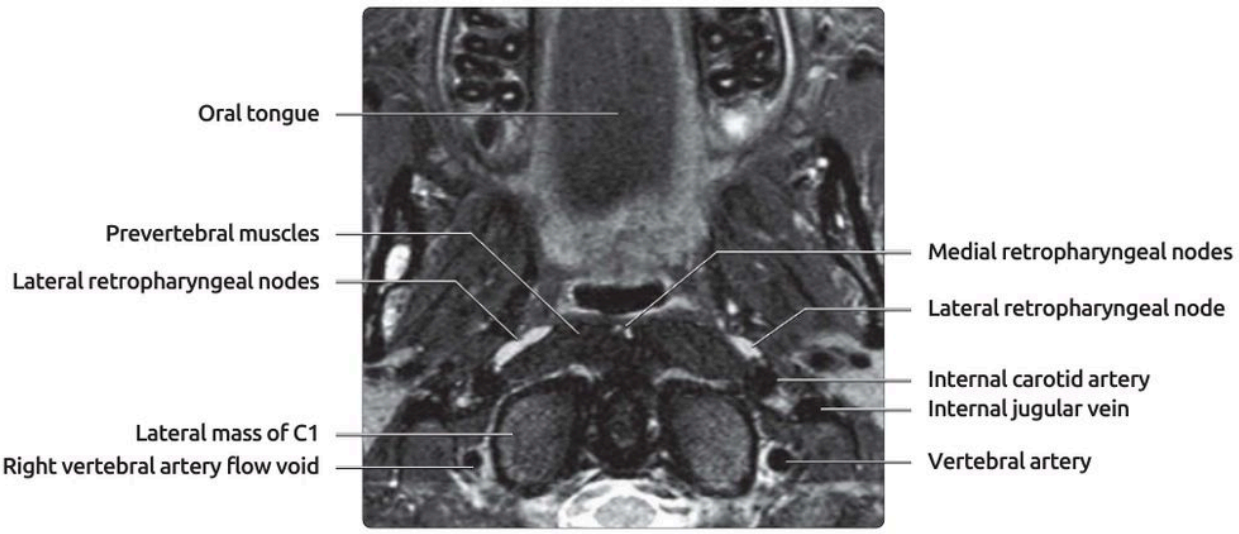


FIGURE 4 Axial fat-suppressed T2: the lateral retropharyngeal nodes (of Rouvière) anterolateral to the prevertebral muscles, and the medial retropharyngeal nodes.
Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 435

CHAPTER TWO

2 Is this node abnormal?

Use all the features together.

2.1 The malignant features

Size alone misclassifies one node in five or more, so use every feature together.

- **Necrosis**: a focal low-density defect in an untreated node is the most reliable sign of metastasis. Do not mistake the normal fatty hilum for necrosis.
- **Shape**: normal nodes are oval or kidney-shaped with a fatty hilum; malignant nodes become **round** (long-to-short axis ratio under 2) and lose the hilum.
- **Size**: thresholds vary; commonly used limits are about 10 mm in short axis (a little more for the jugulodigastric node) and 5 to 8 mm for a lateral retropharyngeal node. A **cluster** of three or more borderline nodes in the drainage area of a tumour is suspicious.
- **Extranodal extension**: an irregular, ill-defined margin, stranding of the surrounding fat, and invasion of muscle, vessel or skin. It changes the stage and the treatment.
- **Asymmetry and enhancement**: a node that enhances more than its neighbours, or that is new, is suspicious.

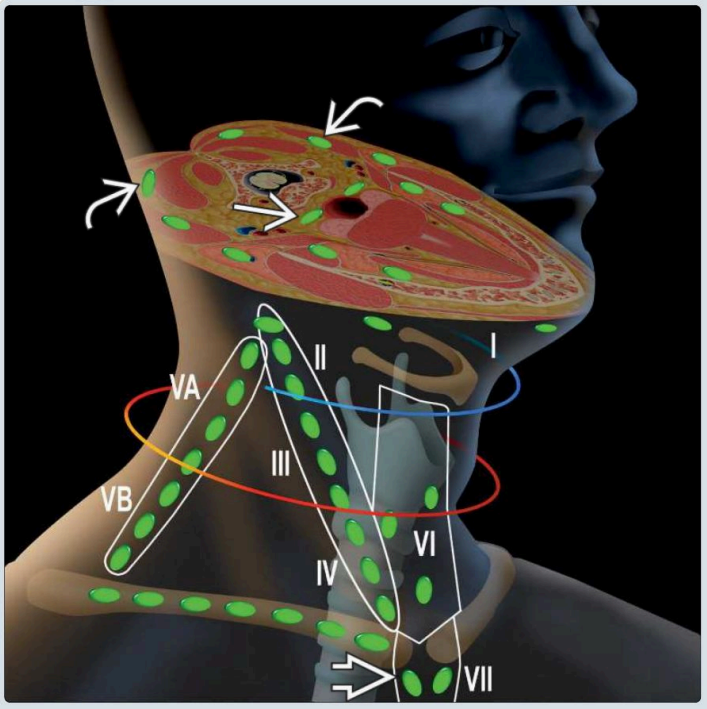


FIGURE 5 The nodal network of the neck, drawn with the levels outlined, the hyoid plane (blue) and cricoid plane (orange-red) that divide them, and, in the axial slice above, the retropharyngeal and superficial nodes (arrows). Level II is the commonest site of metastasis from squamous carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 540

2.2 Ultrasound, MRI and PET

Ultrasound shows the hilum and its vessels best: a normal node is oval with an **echogenic hilum** and **hilar vascularity**; a metastatic node is round, loses the hilum and shows **peripheral** vascularity, and can be sampled by FNA in the same sitting. On **MRI**, metastatic squamous carcinoma has a low ADC on diffusion. **PET-CT** improves detection in advanced disease and the unknown primary, but misses small and cystic nodes and gives false positives with reactive nodes.

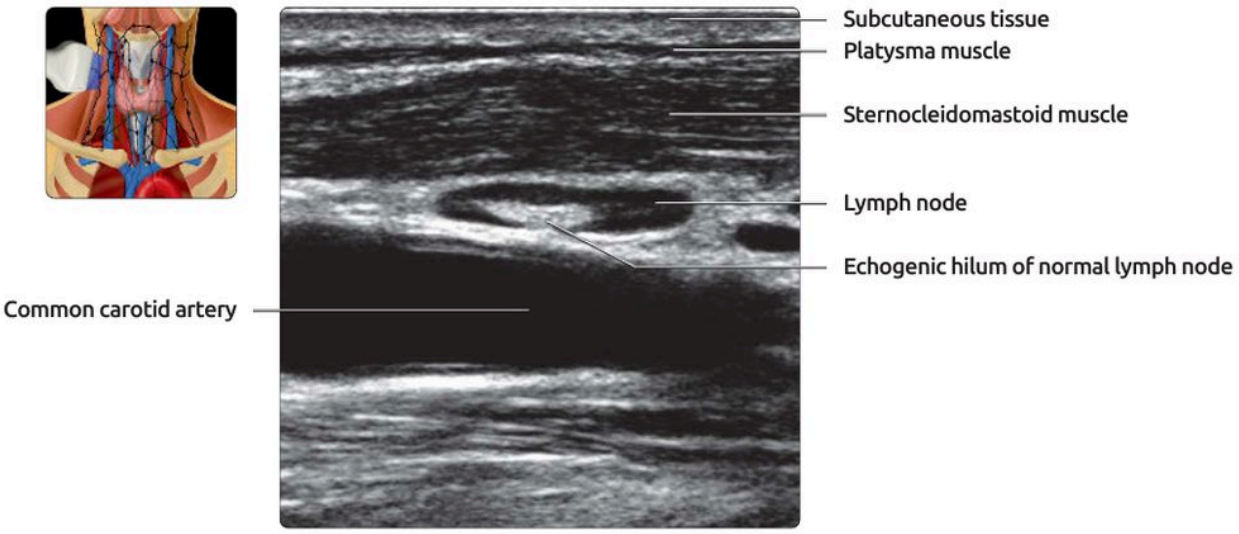


FIGURE 6 Ultrasound of a normal cervical node: an elliptical, hypoechoic node with an echogenic fatty hilum, in front of the common carotid artery.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 436

2.3 Unresectability

A nodal mass that **encases the carotid** (contact of 270° or more), invades the **prevertebral muscles** or the skull base, or infiltrates the skin widely is usually unresectable. Contact under 180° usually means the artery can be peeled free.

3 CHAPTER THREE Where is the primary?

Patterns of spread.

3.1 Reading the nodes backwards

Each primary site drains to predictable levels, so the level of an abnormal node points to the primary. Nodes outside the usual pattern should prompt a search for a second primary.

THE NODE AND THE LIKELY PRIMARY

NODES	LIKELY PRIMARIES
IA and IB	Lip, anterior oral cavity, floor of mouth, skin of the face

NODES	LIKELY PRIMARIES
IIA (jugulodigastric)	Oropharynx (tonsil, tongue base), oral tongue, supraglottis, nasopharynx
Retropharyngeal and VA	Nasopharynx; also oropharynx and hypopharynx
III and IV	Larynx, hypopharynx, thyroid
VI	Thyroid, subglottis, cervical oesophagus
Parotid and preauricular	Skin of the scalp, ear and face (squamous carcinoma, melanoma)
Left supraclavicular (Virchow)	Chest and abdomen

3.2 The cystic node

A cystic or largely necrotic node in an adult is a **metastasis until proven otherwise**. A **thick-walled** necrotic node most often comes from the tonsil or tongue base; a cystic node with a thin wall in a young adult suggests HPV-related oropharyngeal cancer; a cystic node with **calcification**, bright T1 signal (thyroglobulin) or avid enhancement suggests **papillary thyroid carcinoma**, even when the thyroid looks normal.



FIGURE 7 Papillary thyroid carcinoma in a node. Left: contrast CT, a large heterogeneous level III node displacing the jugular vein, with calcification lower down. Right: ultrasound, marked internal heterogeneity; FNA confirmed papillary carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 340

4 CHAPTER FOUR Not every abnormal node is cancer

The common mimics.

4.1 Reactive, suppurative, tuberculous and lymphoma

Reactive nodes are enlarged but oval, with a preserved hilum and no necrosis, often in several levels on both sides. **Suppurative** nodes, usually in children, have a thick enhancing rim around a fluid centre with **inflammatory stranding** of the fat and skin around them; they may progress to an abscess. **Tuberculous** nodes are often matted, necrotic and rim-enhancing, in the posterior triangle and lower neck, with less surrounding inflammation than their necrosis suggests, and calcify later. **Lymphoma** produces large, homogeneous, **non-necrotic** nodes in many chains, often with involvement of Waldeyer's ring, and restricts strongly on diffusion.

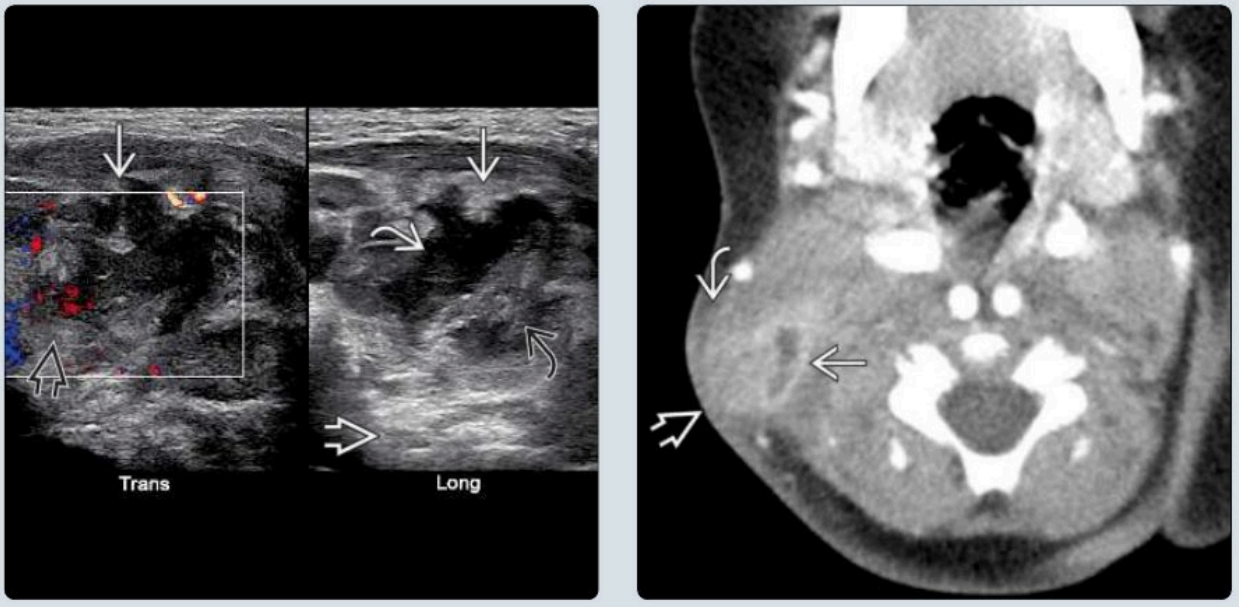


FIGURE 8 Suppurative lymphadenitis. Right: contrast CT, a rim-enhancing, centrally necrotic level II node with cellulitis and oedema of the overlying skin and fat.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 314

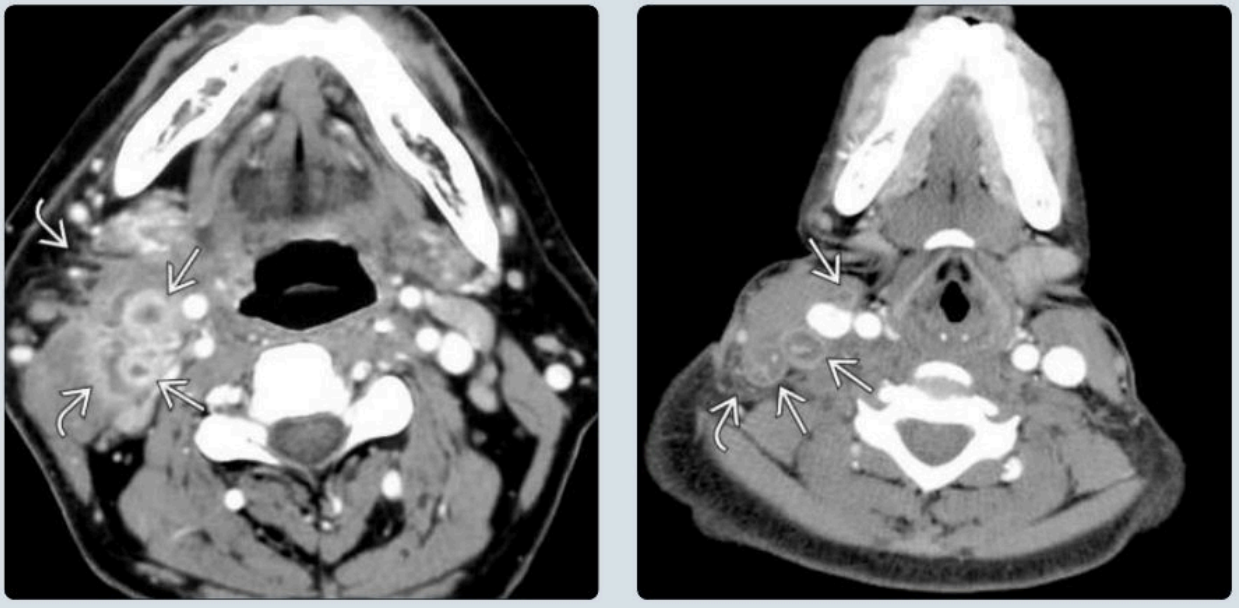


FIGURE 9 Tuberculous lymphadenitis. Left: multiple necrotic, rim-enhancing nodes with extranodal inflammation. Right: matted necrotic nodes that had not responded to treatment.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 318

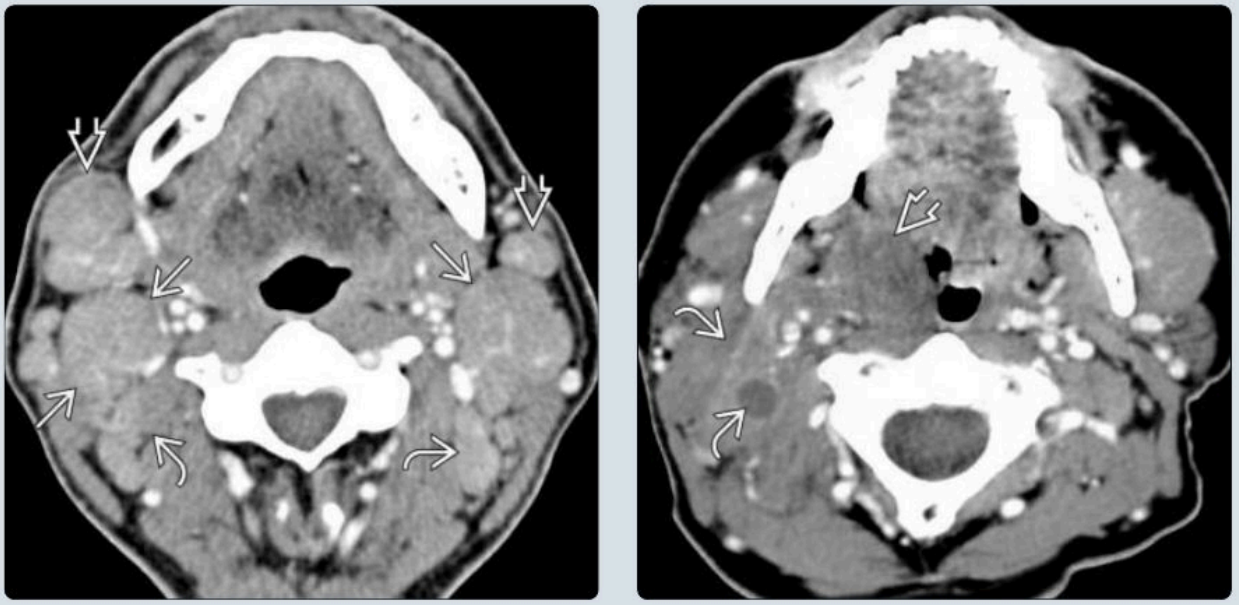


FIGURE 10 Non-Hodgkin lymphoma. Left: large, round, solid nodes in many chains on both sides without necrosis. Right: nodal and tonsillar lymphoma, with a partly necrotic level II mass.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 336

READING THE NODES ON ANY HEAD AND NECK SCAN

1 Every level, both sides

- → I to VII
- → retropharyngeal
- → parotid, facial, occipital, supraclavicular

2 Each suspicious node

- → level
- → short axis
- → shape and hilum
- → necrosis or cystic change
- → calcification
- → margins (extranodal extension)

3 Relations

→ carotid contact in degrees

→ jugular vein

→ prevertebral muscles

→ skin

4 **Pattern**

→ expected drainage of the primary

→ contralateral nodes

→ nodes outside the pattern

5 **Next step**

→ ultrasound and FNA

→ PET-CT for unknown primary or advanced disease

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Imaging levels** and their landmarks: hyoid, cricoid, submandibular gland, sternocleidomastoid, jugular vein, manubrium.
- **Retropharyngeal nodes** and the nasopharynx.
- **Malignant features:** necrosis, shape, size, extranodal extension.
- **Ultrasound** features of normal and abnormal nodes.
- **Node to primary** patterns; the cystic node in an adult.
- **Carotid encasement** and unresectability.

- **Mimics:** reactive, suppurative, tuberculous, lymphoma.

5.2 Common pitfalls

COMMON PITFALLS

- Using size alone to call a node benign or malignant.
- Calling the fatty hilum necrosis.
- Forgetting the retropharyngeal nodes.
- Calling a cystic level II mass in an adult a branchial cyst.
- Missing a thyroid primary behind a calcified or cystic node.

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